1 Present one task trial

Instructions + one solved example

Target sequence

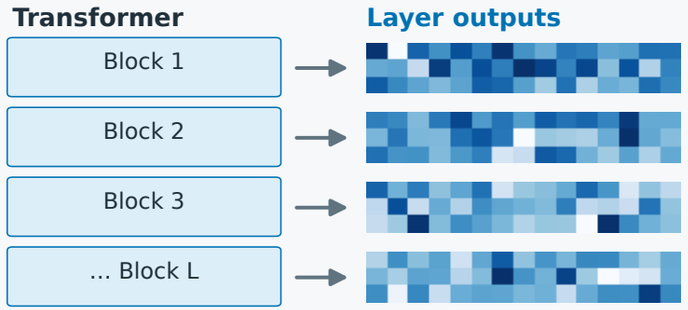
star star star guitar star star star ?

Options: truck / star / cube / guitar

Tokenize the full prompt

400 independent trials per model
No previous-trial context

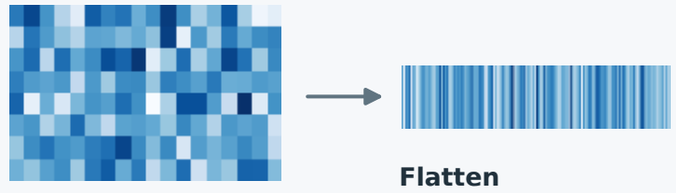
2 Read internal activations



Retain the target-sequence token span
Hidden features at every layer

3 Build trial representations

Tokens × hidden features



One vector per trial, per layer

Keep token-position information
Stack trial vectors for each layer

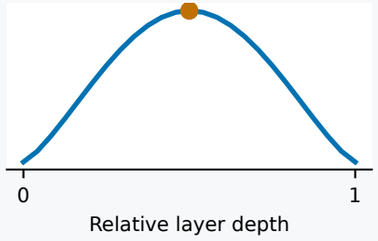
Repeat across layers → choose a layer → compare pattern geometry

4 Select a task-informative layer

Task-design RDM

Same pattern: 0
Different pattern: 1

RSA at each layer



Correlate 400 × 400 trial RDMs with the task-design RDM; retain the best layer.

Layer selection uses no human EEG.

5 Construct the pattern RDM

8 pattern-mean vectors

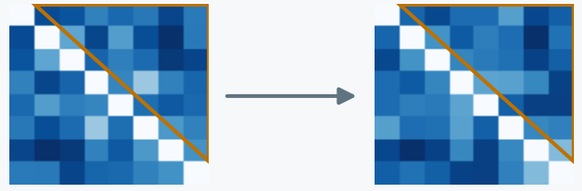
8 × 8 RDM

Average trial vectors within each pattern at the selected layer.

Dissimilarity = 1 − Pearson r
Repeat separately for each model.

6 Compare with human geometry

LLM **Human EEG**



Extract the same 28 unique pattern pairs

Pearson correlation between RDM entries

One RSA value per model, EEG measure and regional group.

Answers are scored separately for behavioral accuracy; the RSA features are internal activations.

Schematic illustration: token grids, matrices and layer trajectory are not measured data. Eight models are analyzed separately.